

EE 5239 Nonlinear Optimization

Mathematical Background

Mingyi Hong

University Of Minnesota

Outline

1 Mathematical Review

Overview

- 1 Notations: Sets, functions, derivatives, gradients
- 2 Vectors, matrices
- 3 Norms, sequences, limits, continuity
- 4 Mean value theorems
- 5 Implicit function theorem
- 6 Contraction mappings
- 7 Reference Appendix A, B of the textbook
- 8 Get yourself familiar with them

Notations

① **Sets:** $X, x \in X, X_1 \cup X_2, X_1 \cap X_2$

② **Inf and Sup:**

The supremum of a nonempty set $X \subset \mathbb{R}$ is the **smallest** scalar y :

$$y \geq x, \forall x \in X$$

The infimum of a nonempty set $X \subset \mathbb{R}$ is the **largest** scalar y :

$$y \leq x, \forall x \in X$$

If $\sup X \in X$ (or, $\inf \in X$), then we say $\sup X = \max X$ (or, $\inf X = \min X$).

$$\sup\{1/n \mid n \geq 1\} = ?, \quad \inf\{x \in \mathbb{R} \mid 0 < x < 1\} = ?$$

③ **Function:**

$f : X \rightarrow Y$, X is called the domain, Y is called the range

Monotonicity $\longrightarrow$ **Inverse function** f^{-1} exists

Vectors

- ① **Vector:** a vector $\mathbf{x} = [x_1, \dots, x_n]$ is a **column** of scalars
- ② **Linear combination:** if $\mathbf{x} = [x_1, \dots, x_n]$ and $\mathbf{y} = [y_1, \dots, y_n]$, then the linear combination is given by

$$\alpha \mathbf{x} + \beta \mathbf{y} = (\alpha x_1 + \beta y_1, \alpha x_2 + \beta y_2, \dots, \alpha x_n + \beta y_n)$$

- ③ **Inner product:** $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}' \mathbf{y} = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$

Question: when is inner product positive, negative, zero?

Orthogonality: $\mathbf{x} \perp \mathbf{y}$ iff (if and only if) $\langle \mathbf{x}, \mathbf{y} \rangle = 0$.

- ④ **Linearly Independent:** A set of vectors $\{\mathbf{x}^1, \dots, \mathbf{x}^r\}$ are **linearly independent** if there does not exist a $(\alpha_1, \dots, \alpha_r) \neq 0$ s.t.

$$\alpha_1 \mathbf{x}^1 + \alpha_2 \mathbf{x}^2 + \dots, \alpha_r \mathbf{x}^r = 0.$$

Vectors

- ① Basis and dimension of a subspace;
Orthogonal complement of a subspace S :

$$S^\perp := \{\mathbf{x} \mid \langle \mathbf{x}, \mathbf{y} \rangle = 0, \forall \mathbf{y} \in S\}$$

- ② **Vector norms:** A norm $\|\mathbf{x}\|$ on $\mathbb{R}^n$ that assigns a **scalar** $\|\mathbf{x}\|$ to every $\mathbf{x} \in \mathbb{R}^n$ that satisfying

- ① $\|\mathbf{x}\| \geq 0$ for all $\mathbf{x}$
- ② $\|c\mathbf{x}\| = |c|\|\mathbf{x}\|$ for all $c \in \mathbb{R}$ and all $\mathbf{x}$
- ③ $\|\mathbf{x}\| = 0$ iff $\mathbf{x} = 0$
- ④ $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y}$

- ③ Measure some kind of “distance”

Vectors

1 Common norms

$$\text{Euclidean Norm : } \|\mathbf{x}\|_2 = (\mathbf{x}'\mathbf{x})^{1/2} = \left(\sum_{i=1}^n |x_i|^2 \right)^{1/2}$$

$$\ell_p \text{ Norm : } \|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, \text{ for some } p \geq 1$$

$$\ell_1 \text{ Norm : } \|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$$

$$\ell_\infty \text{ Norm : } \|\mathbf{x}\|_\infty = \max_i |x_i|$$

2 Cauchy-Schwartz inequality: $\langle \mathbf{x}, \mathbf{y} \rangle \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$

Cauchy-Schwartz inequality

An important inequality about the inner product of two vectors is the **Cauchy-Swartz inequality**

- 1 Characterizes the inner product of two vectors with their norms
- 2 Given two vectors $\mathbf{x}$ and $\mathbf{y}$ of the same size, we have

$$\langle \mathbf{x}, \mathbf{y} \rangle \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$$

- 3 Why this is true? Geometrically?
- 4 Useful fact about inner product

$$\langle \mathbf{x}, \mathbf{y} \rangle = \cos(\theta) \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$$

where θ is the **angle** between $\mathbf{x}$ and $\mathbf{y}$

Derivatives

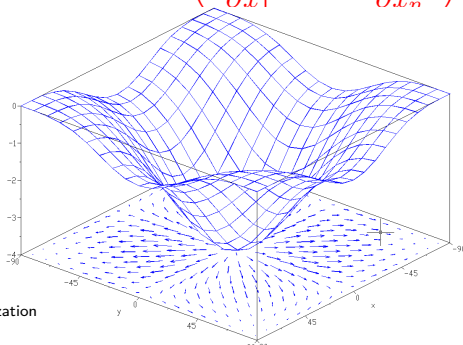
Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuously twice differentiable function.

- ① Derivative (where $\mathbf{e}_i$ is the i th unit vector of $\mathbb{R}^n$)

$$\frac{\partial f(\mathbf{x})}{\partial x_i} := \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{e}_i) - f(\mathbf{x})}{t}$$

- ② Gradient vector (a column vector) [example: $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x} - \mathbf{b}$]:

$$\nabla f(\mathbf{x}) = \left(\frac{\partial f(\mathbf{x})}{\partial x_1}, \dots, \frac{\partial f(\mathbf{x})}{\partial x_n} \right)$$



Derivatives

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuously twice differentiable function.

① Hessian matrix:

$$\nabla^2 f = \left[\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j} \right] \in \mathbb{R}^{n \times n}$$

② Taylor expansion:

$$f(\mathbf{y}) - f(\mathbf{x}) = \nabla f(\mathbf{x})'(\mathbf{y} - \mathbf{x}) + \frac{1}{2}(\mathbf{x} - \mathbf{y})' \nabla^2 f(\mathbf{x})(\mathbf{x} - \mathbf{y}) + o(\|\mathbf{x} - \mathbf{y}\|^2)$$

Derivatives

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuously twice differentiable function.

- ① Chain rule: $f : \mathbb{R}^k \rightarrow \mathbb{R}^m$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}^n$, let $h(\mathbf{x}) = g(f(\mathbf{x}))$ (function composition), then

$$\nabla h(\mathbf{x}) = \nabla f(\mathbf{x}) \nabla g(f(\mathbf{x}))$$

For example, we have ($g(z) = z^2$, $f(\mathbf{x}) = \mathbf{Ax} - \mathbf{b}$)

$$\nabla \left(\frac{1}{2} \|\mathbf{Ax} - \mathbf{b}\|^2 \right) = \mathbf{A}'(\mathbf{Ax} - \mathbf{b}), \quad \nabla^2 f(\mathbf{Ax}) = \mathbf{A}'\mathbf{A}$$

Matrices

- ① For any matrix $\mathbf{A}$, we use a_{ij} (or A_{ij}) to denote its (i, j) th entry.
- ② Matrix addition, multiplication, transpose, symmetrix matrices $\mathbf{A} = \mathbf{A}'$.

$$[\mathbf{AB}]' = \mathbf{B}'\mathbf{A}', \mathbf{AB} \neq \mathbf{BA}$$

- ③ Let $\mathbf{A}$ be a $m \times n$ matrix.
 - Range of $\mathbf{A}$: $R(\mathbf{A}) = \{\mathbf{Ax}, | \mathbf{x} \in \mathbb{R}^n\}$;
 - Null space of $\mathbf{A}$: $N(\mathbf{A}) = \{\mathbf{x} | \mathbf{Ax} = 0\}$
 - Rank of $\mathbf{A}$ $\text{Rank}(\mathbf{A})$. Full rank matrix $\mathbf{A}$:
 $\text{Rank}(\mathbf{A}) = \min\{m, n\}$.

- ④ **Inner product:**

$$\langle \mathbf{A}, \mathbf{B} \rangle = \text{Tr}(\mathbf{AB}') = \sum_{i,j} A_{ij}B_{ij}$$

where the trace operate is given by

$$\text{Tr}[\mathbf{A}] = \sum_{i=1}^n A_{ii}$$

Square Matrices

- ① Square matrix ($m = n$); Identity matrix $\mathbf{I}$
- ② Determinant $\det(\mathbf{A})$, inverse $\mathbf{A}^{-1}$. $\mathbf{A}^{-1}$ exists iff $\det(\mathbf{A}) \neq 0$
- ③ Useful identities: $\det(\mathbf{A}) = \det(\mathbf{A}')$
- ④ Orthogonal matrices: $\mathbf{A}\mathbf{A}' = \mathbf{I}$
- ⑤ (Complex) Eigenvalue λ : $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ for some $\mathbf{x} \neq 0$
- ⑥ Spectral radius: $\rho(\mathbf{A}) = \max_i \{|\lambda_i|\}$, λ_i is an eigenvalue of $\mathbf{A}$.

Square Matrices

- ① Eigen-decomposition of a symmetric matrix:

$$\mathbf{A} = \mathbf{P}'\mathbf{\Lambda}\mathbf{P}$$

where $\mathbf{P}$ is orthonormal ($\mathbf{P}'\mathbf{P} = \mathbf{I}$), $\mathbf{\Lambda}$ is diagonal and real.

- ② Positive (semi-) definite matrix: $\mathbf{A} \succeq 0$

$$\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0, \forall \mathbf{x} \quad (0.1)$$

- ③ Property: $\mathbf{A} \succeq 0, \mathbf{B} \succeq 0 \rightarrow \mathbf{A} + \mathbf{B} \succeq 0$; $\mathbf{A} \succeq \mathbf{B} \rightarrow \mathbf{A} - \mathbf{B} \succeq 0$

All eigenvalues of $\mathbf{A}$ are non-negative

- ④ Square root $\mathbf{A}^{1/2}$: $\mathbf{A}^{1/2} := \mathbf{P}\sqrt{\mathbf{\Lambda}}\mathbf{P}$, where $\mathbf{A} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}'$ is the eigen-decomposition of $\mathbf{A} \succeq 0$.

- ⑤ A “generalization” of **positive number** for the scalars

Single Value Decomposition

- ① Single value decomposition (SVD):

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}' \in \mathbb{R}^{m \times n}, \quad (0.2)$$

- $\mathbf{\Sigma} \in \mathbb{R}^{m \times n}$ rectangular diagonal matrix with diagonals $\sigma_1 \geq \sigma_2 \geq \dots, \sigma_n \geq 0$;
- $\mathbf{U} \in \mathbb{R}^{m \times m}$, $\mathbf{V} \in \mathbb{R}^{n \times n}$ orthonormal

- ② Relationship of SVD and ED: σ_i^2 is an eigenvalue of $\mathbf{A}\mathbf{A}'$ (Why?)

$$\mathbf{A}\mathbf{A}' = (\mathbf{U}\mathbf{\Sigma}\mathbf{V}')(\mathbf{V}\mathbf{\Sigma}\mathbf{U}') = \mathbf{U}\mathbf{\Sigma}^2\mathbf{U}'$$

- ③ Condition number: $\kappa(\mathbf{A}) = \sigma_1/\sigma_n$

Matrices and Norms

1 Norms:

$$\text{Frobenius Norm : } \|\mathbf{A}\|_F = \left(\sum_{i,j} |A_{ij}|^2 \right)^{1/2} = \left(\sum_i \sigma_i^2 \right)^{1/2} \quad (0.3)$$

$$\text{Nuclear Norm : } \|\mathbf{A}\|_* = \sum_i \sigma_i \quad (0.4)$$

$$\text{Matrix 2-norm : } \|\mathbf{A}\|_2 = \sup_{\mathbf{x} \neq 0} \frac{\|\mathbf{Ax}\|_2}{\|\mathbf{x}\|_2} = \max_i \sigma_i \quad (0.5)$$

2 Useful inequalities:

$$\|\mathbf{Ax}\|_2 \leq \|\mathbf{A}\|_2 \|\mathbf{x}\|_2, \quad \|\mathbf{A}\|_* \geq \|\mathbf{A}\|_F \geq \|\mathbf{A}\|_2$$

Contraction Mapping

- ① **Lipschitzian Property:** $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfies

$$\|f(\mathbf{x}) - f(\mathbf{y})\| \leq \gamma \|\mathbf{x} - \mathbf{y}\|, \forall \mathbf{x}, \mathbf{y}$$

γ is called the **Lipschitz constant**; the function is called **Lipschitz continuous**

- ② If $\gamma \leq 1$, then f is called a **non-expansive mapping**
- ③ If $\gamma < 1$, then f is called a **contraction mapping**
- ④ **Fixed point theorem:** If f is contraction, then the iterated function sequence

$$\mathbf{x}, f(\mathbf{x}), f(f(\mathbf{x})), \dots,$$

converges to a unique fixed point $\mathbf{x}^*$ (independent of $\mathbf{x}$) satisfying

$$\mathbf{x}^* = f(\mathbf{x}^*).$$

Homework

- ① Getting yourself familiar with mathematical backgrounds
- ② **Reading:** Appendix A of the textbook.